



Graphene and 2D Materials: Science, Processing, and Applications (ChE 494)

Homework 1: Science of 2D Materials

September 10, 2019

Due Date: September 17, 2019

Total: 100

Submit Your Homework by September 17, 2019 at 4:00 pm in EIB 142 or by E-Mail to sbehura1@uic.edu

- Graphene is a two-dimensional (2D) crystal with carbon atoms arranged in a honeycomb structure as shown in Figure 1. Using the above information and Figure 1, answer the following questions.

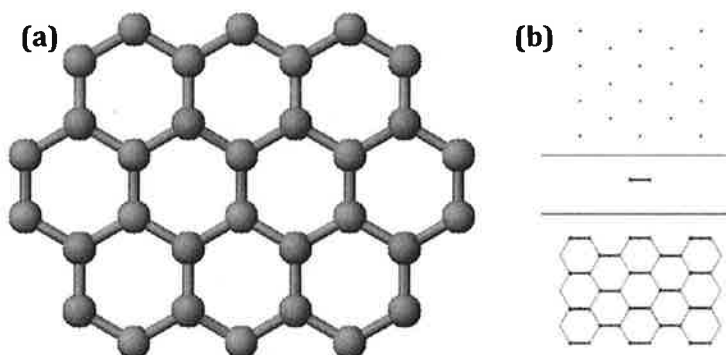


Figure 1: Graphene Lattice.

- Assuming that the unit cell is a hexagon of carbon atoms (Figure 1 (a)), how many atoms of carbon are there in each unit cell (Explain)?
 - Crystal Structure = Lattice + Basis. In Figure 1 (b), designate: (i) lattice and its type, (ii) basis and how many atoms in basis, and crystal structure.
- What is Stone-Wales (SW) defect in graphene and state SW defect's consequences on the properties of graphene?
 - Intrinsic monolayer graphene is a zero bandgap ($E_g = 0$ eV) semiconductor. Show the band structure (Energy-Momentum Linear Dispersion) of an intrinsic monolayer graphene. Suggest some techniques (minimum two) to open bandgap in graphene.
 - Calculate the lattice unit vectors (a_1 and a_2) in the following honeycomb graphene crystal lattice.

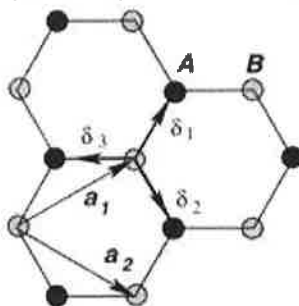
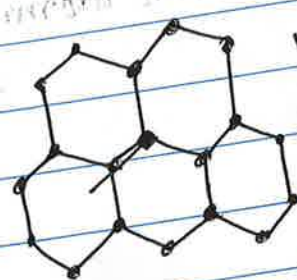


Figure 2: Graphene Lattice.

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① @ Unit cell of graphene lattice is a hexagon of carbon atoms. Each point of a hexagonal unit cell contains an carbon atom with which is shared by the two other unit cells as well. So, each of the six points in the hexagonal unit contains $\frac{1}{3}$ of an carbon atom. So the number of atoms in a hexagonal unit cell is

$$\frac{1}{3} \times 6 = 2 \text{ atoms}$$



Crystal structure of Graphene.



Fig: Hexagonal unit cell of Graphene.

② The carbon atoms in graphene is condensed in a honeycomb lattice with sp^2 hybridization. This can be defined as a triangular Bravais lattice with a two-atom unit cell, and can be regarded as two interlocking triangular lattices.

[Hexagonal lattice]

This is lattice

① Lattice + Basis = Crystal.

In the Figure,

There are two atoms in the Basis.

Hexagonal Lattice

In the crystal structure $\Rightarrow$ the total number of

Carbon atom is $13 \times \frac{1}{3} \times 6 = 26$

Basis.

Crystal structure Graphene

② Stone-Wales (SW) effect in Graphene: $\rightarrow$ A Stone-Wales defect is a crystallographic defect that changes the connectivity of two π -bonded carbon atoms by 90° rotation of atoms with respect to the midpoint of their bond. In graphene, the SW effect transforms four hexagons into two pentagons and two heptagons [SW (55-77) defect].

The defect has a formation of energy $E_f \approx 5 \text{ eV}$.

When the transformation occurs via the simultaneous movement of in plane bond rotation of involved atoms their kinetic barrier is 10 eV. The defect structure has no dangling bond and has the same number of atoms as the original graphene. The reverse transformation energy barrier is

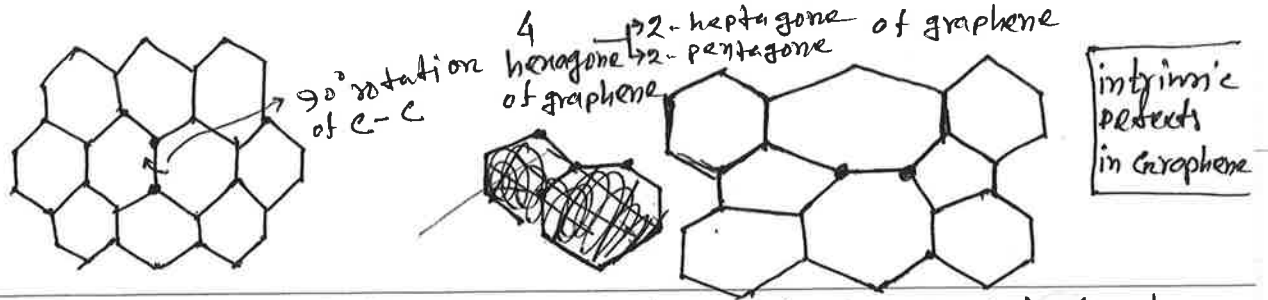


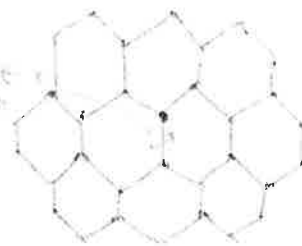
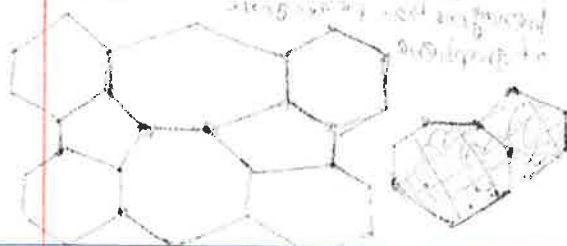
Fig: Stone-Wales defect (SW) in Graphene.

The SW defects might formed under the effect of irradiation or during quenching from high temperature at the ~~synthesis~~ synthesis storage and remain stable under normal conditions. It is a point defect.

The consequences of SW defect in graphene is it ~~is~~ violates the lattice symmetry and induced the dielectric gap in graphene, which has application in nanoelectronics. It is a potential absorbant ~~at~~ center for hydrogen and some chemical elements, which also has applications.

It is partially responsible for roughness of graphene. The SW effects lead to a strong bending of monolayer and it can promote the formation of nonplanar nanostructures like fullerenes and nanotubes. ^{SW} Defects decrease the thermal conductivity of graphene.

graphene



③ Intrinsic monolayer graphene is a zero bandgap ($E_g = 0$ eV) semiconductor. The band structure of an intrinsic monolayer graphene is below:→

mass of graphene is $m = 0$

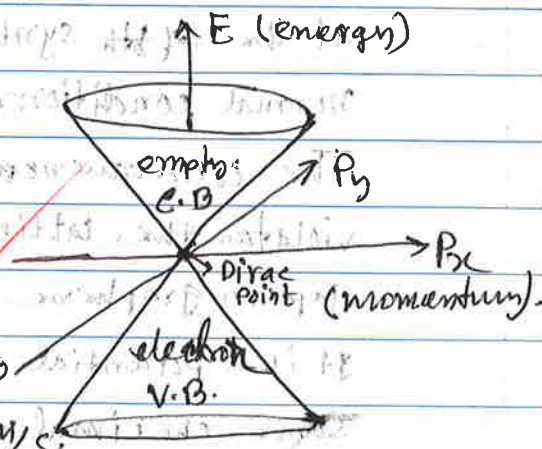
$$E = m^2 c^4 + p^2 c^2$$

$$E = \pm p c \Rightarrow E = p c$$

for graphene, $E = v_F p$

$$E = \pm v_F |\vec{p}| ; v_F = \text{fermi velocity}$$

$$v_F \sim \frac{1}{300} c \approx 3 \times 10^8 / 300 = 10^6 \text{ m/s}$$



④ To open band gap in graphene following techniques can be used:→

① Substrate Induced band gap opening:→ Growth of epitaxial graphene on epitaxial h-BN induces band gap in graphene. (53 meV).

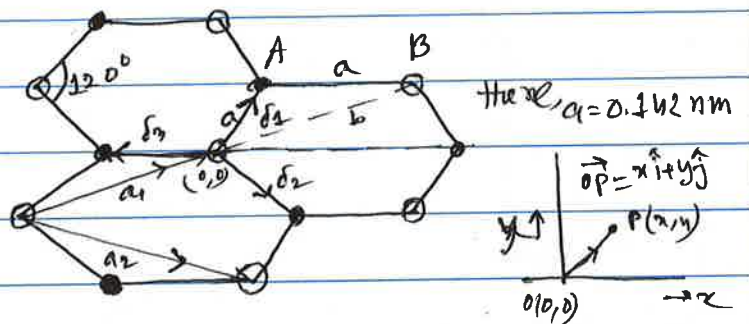
② Chemical Substitution doping: Doping of N_2 can convert the graphene to n-type semiconductor while Boron doping can produce p-type graphene semiconductor ~~for graphene~~.

③ Hybrids:→ Doping of graphitic carbon lattice with both boron and N_2 simultaneously to form BCN semiconductors or $B_x C_y N_z$ nanotubes.

④ Quantum Confinement:→ The above mentioned method have drawbacks like low charge carriers mobility which can be solved by making quasi 1-D

nanoribbon from 2-D larger area planar graphene sheet, which will confine electron from 2-D to 1-D and expanding the electronic structure thereafter. It induces a band gap in graphene with no impurities and enough charge carrier mobility.

- (4) The crystal structure of graphene is shown in the picture. The hexagonal lattice are the resultant of two interpenetrating triangular lattices, i.e. sublattices. The real space translational vectors are



$$R = n_1 a_1 + n_2 a_2$$

The three nearest neighbour vectors are d_1, d_2, d_3 .

Now, $a_1 = d_1 - d_3$, $a_2 = d_2 - d_3$

now, position of

$$\begin{aligned} d_3 &= (-a, 0), & d_1 &= (a \cos 60^\circ, a \sin 60^\circ), & d_2 &= (a \cos 120^\circ, -a \sin 120^\circ) \\ &= (-a, 0), & &= \left(\frac{a}{2}, \frac{\sqrt{3}a}{2}\right), & &= \left(\frac{a}{2}, -\frac{\sqrt{3}a}{2}\right) \end{aligned}$$

$$\therefore a_1 = \left(\frac{3a}{2}, \frac{\sqrt{3}a}{2}\right), \quad a_2 = \left(\frac{3a}{2}, -\frac{\sqrt{3}a}{2}\right).$$

$$= \frac{a}{2} (3, \sqrt{3})$$

$$a_2 = \frac{a}{2} (3, -\sqrt{3}).$$

Ans

Ans

